

IT457

Cloud Computing

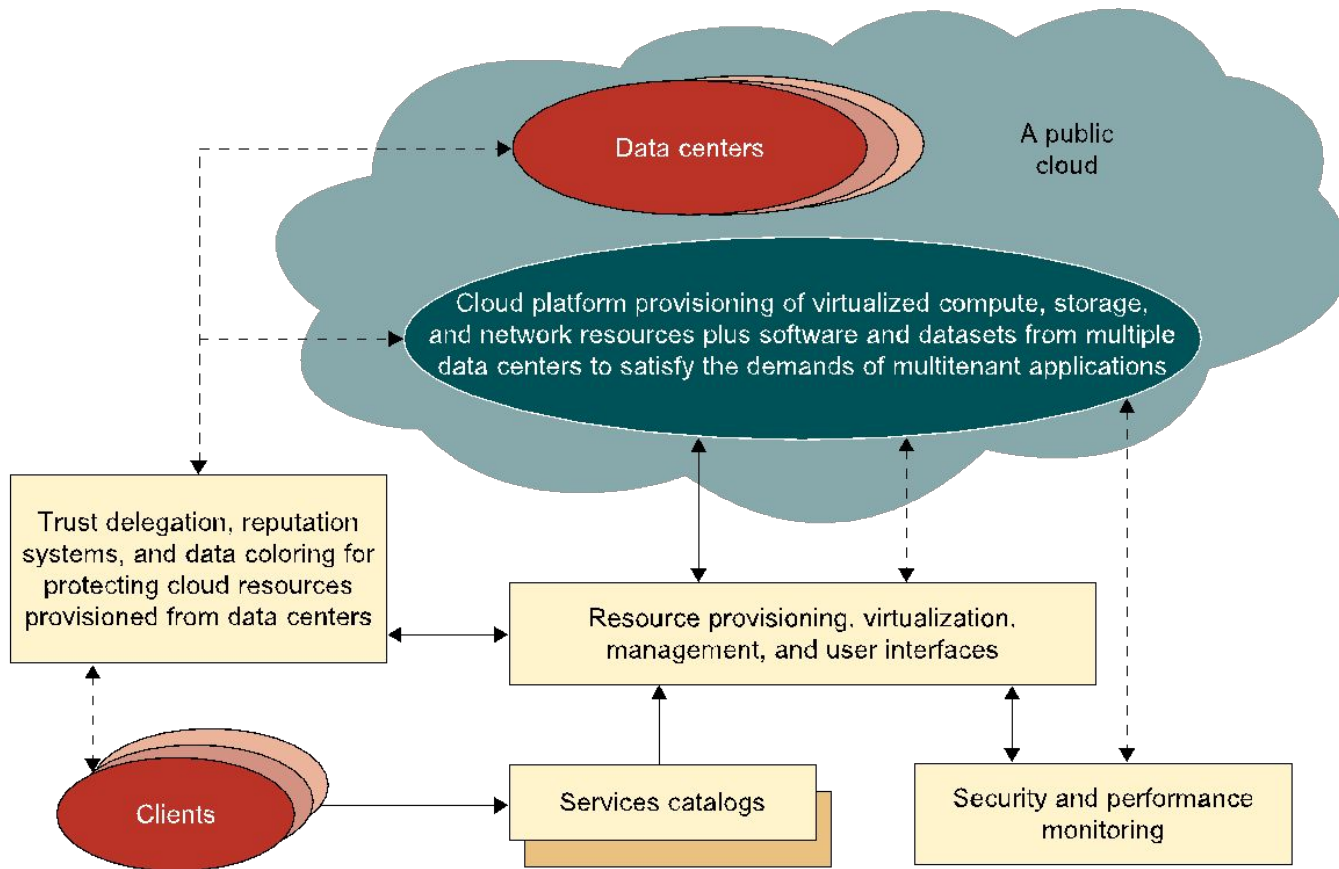
Week 6_ Lec 1

Cloud arch

Design Goal for Cloud Platform

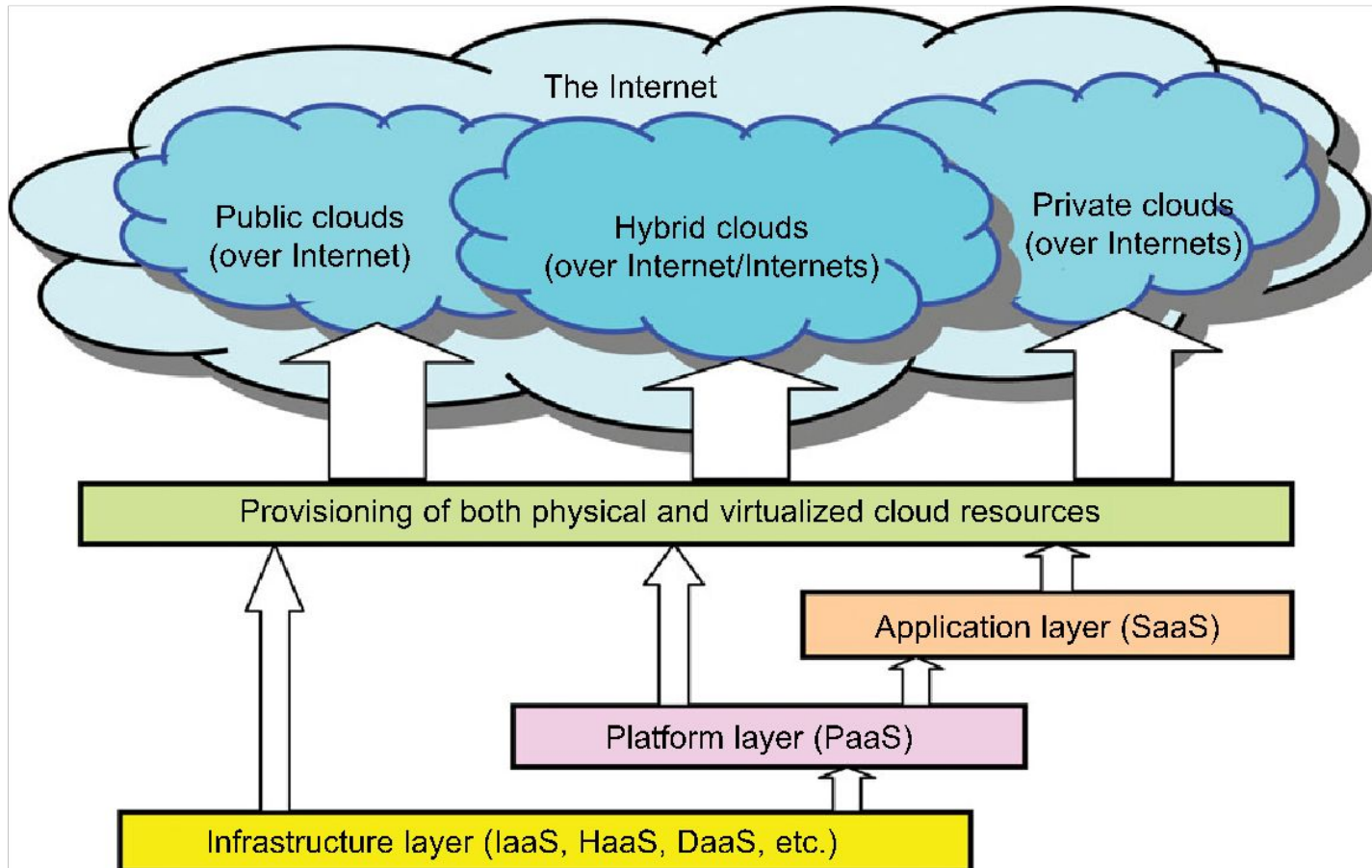
- Scalability, virtualization, efficiency, and reliability
- Cloud management receives
 - the user request
 - finds the correct resources
 - Calls the provisioning services
 - Invoke resources in the cloud.
 - The cloud management software needs to support both physical and virtual machines.

A generic Security Aware Cloud Arch



- Dynamic provisioning/ deprovisioning of resources
- Distributed storage
- Framework for processing large data
 - Distributed File system
- SAN, DBs, Firewalls
- Web services
- Monitoring / metering units
 - Track usage and performance of provisioned resources
- Private clouds – easier to manage
- Public clouds – easier to access

Layered Cloud Arch

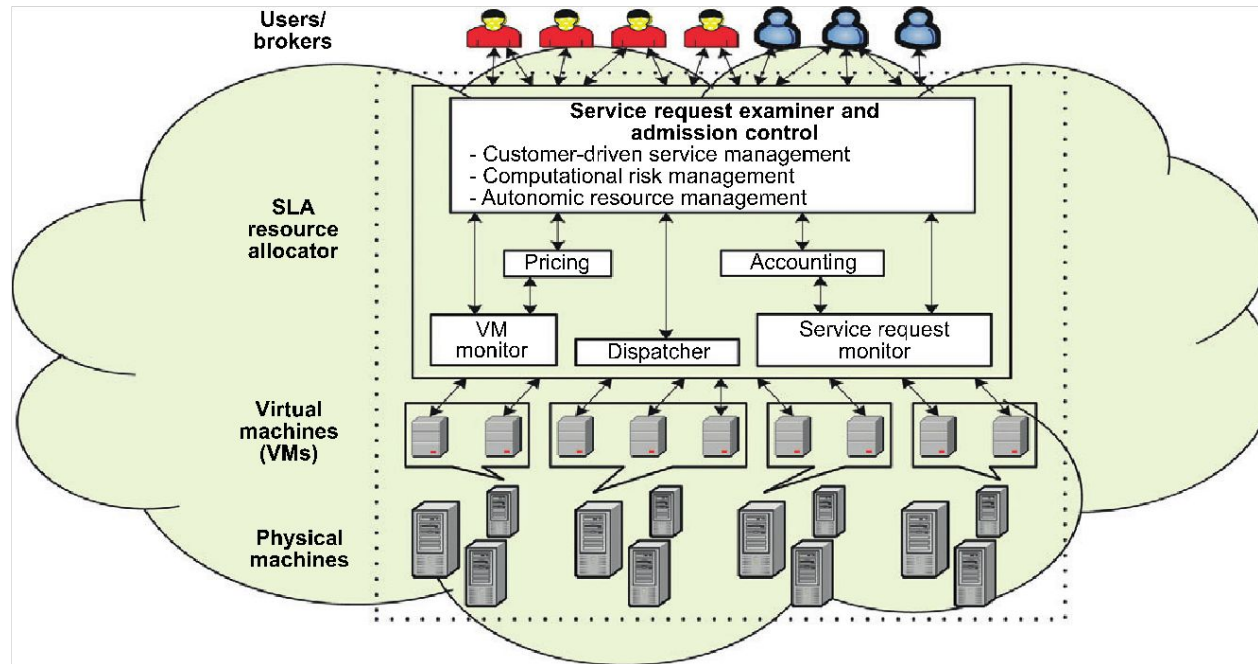


- Infrastructure
 - HW and Virtualized HW
 - Support IaaS
- Platform
 - Collection of SW resources
 - Implement SaaS
- Application

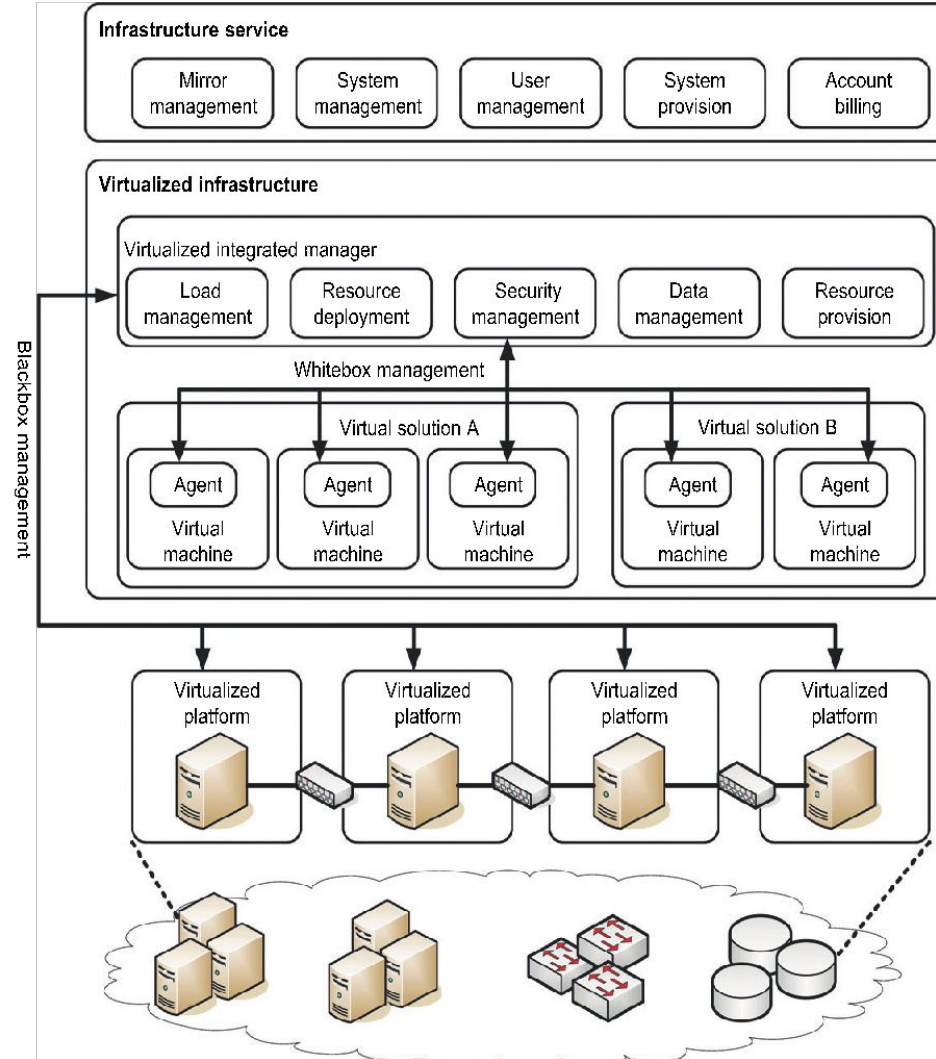
- Amazon EC2 provides
 - Virtualized CPU resources to users
 - Management of these provisioned resources.
- Salesforce.com
 - Services at the application layer demand more work from providers.
 - CRM service
 - hardware at the bottom layer and the software at the top layer, but also the platform and software
 - tools for user application development and monitoring.

Market Oriented Cloud Arch

Market-oriented cloud architecture is a system design that manages and allocates cloud computing resources based on economic models, supply and demand, and explicit service agreements



Infrastructure reqm for Virtualization



Architectural design Challenges

- Challenge 1—Service Availability and Data Lock-in Problem
- Challenge 2—Data Privacy and Security Concerns
- Challenge 3—Unpredictable Performance and Bottlenecks
- Challenge 4—Distributed Storage and Widespread Software Bugs
- Challenge 5—Cloud Scalability, Interoperability, and Standardization
- Challenge 6—Software Licensing and Reputation Sharing

N-Tier

- Traditional layered
- Higher layer calls lower ones, not vice-versa
- Can be a liability – not easy to update individual components
- Suitable for app migration
- Generally IaaS

Web-Queue-Worker

- Purely PaaS
- Web FE + Worker BE
- FE <-> BE communication through async msg Q
- Suitable for simple domains

Microservices

- Many small independent services
- Services are loosely coupled
- Communicate through API contracts

Event-driven Architecture

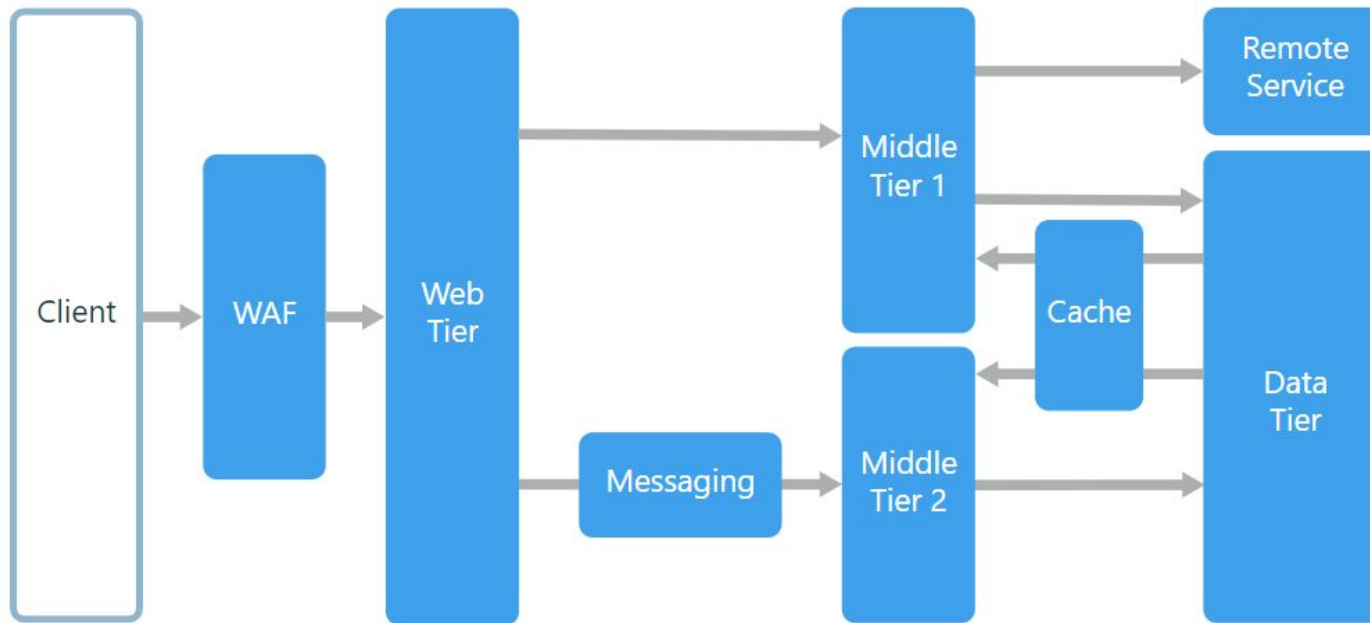
- Uses pub-sub model
- Pubs & Subs are independent
- Suitable for apps that ingest large volume of data
- Suitable when subsystems need to perform different actions on same data

Big Data Big Compute

- Specialized architecture that fit certain profiles
- Divides large datasets into chunks, parallel process and analysis

N-tier Architecture Style

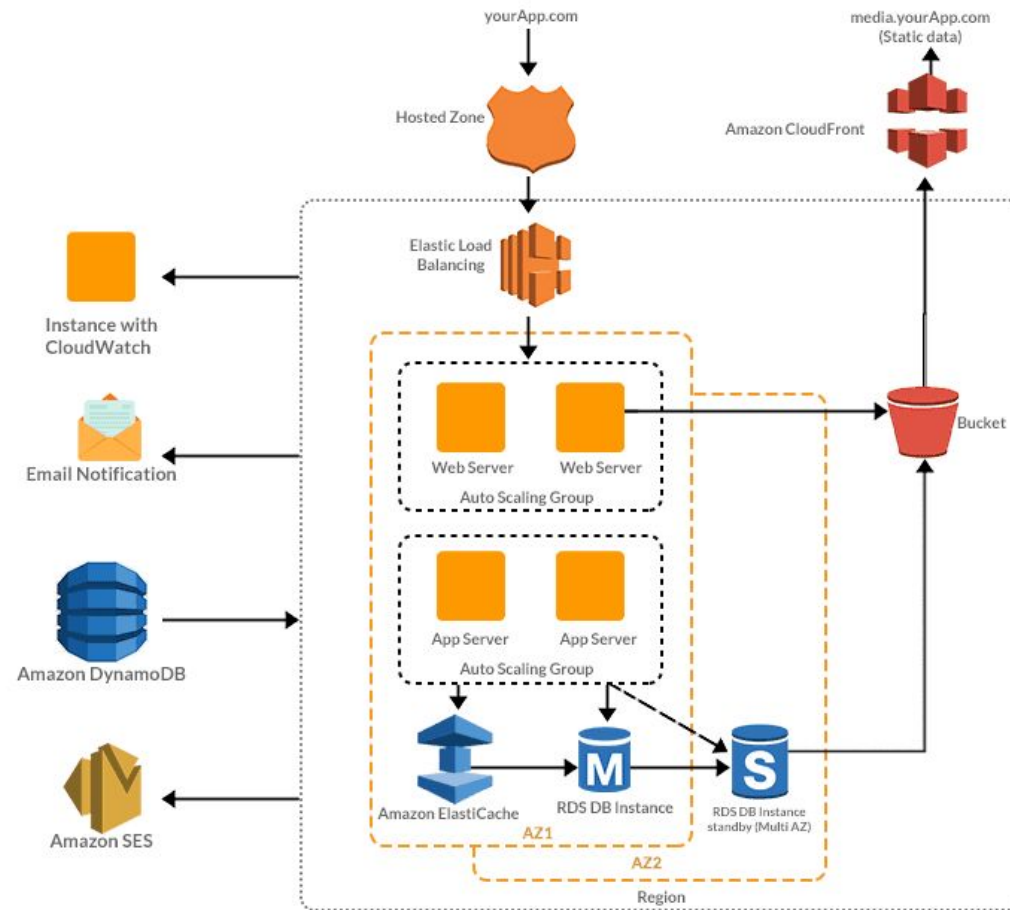
- An N-tier architecture divides an application into **logical layers** and **physical tiers**.
- Layers are a way to separate responsibilities and manage dependencies.
- Tiers are physically separated, running on separate machines.



When to use this architecture

- Typically implemented as infrastructure-as-service (IaaS) applications
 - Each tier running on a separate set of VMs.
- Consider an N-tier architecture for:
 - Simple web applications.
 - Migrating an on-premises application to any cloud platform with minimal refactoring.
 - Unified development of on-premises and cloud applications.

AWS Architecture



- Web Tier: It contains web servers like Apache.
- App Tier: It contains app servers.
- Database Tier: It contains database services like Amazon RDS (Relational Database Service).

Role of Traffic Manager in Multi-region deployment

- Allows users to control traffic distribution across various application endpoints.
- Distributes the traffic using the configured routing method
 - Priority, Weighted, Performance, Geographic
- Constant monitoring of the endpoint health.
- Automatic failover in case the endpoints fail.
- Functions at the DNS level.
 - Uses the DNS to route clients to precise service endpoints, according to the traffic-routing rules and methods.
- Traffic Manager operation modes:
 - Active | Passive – hot failover
 - Active | Passive – cold failover
 - Active | Active – load balanced

Three Tier Layers

- Presentation /Web tiers: UI, browser using HTML, CSS, JS
- Application/Logic -Business logic, Python, Java
- Data/DB: RDBMS MySQL, PostgreSQL / NoSQL MongoDB,

- Network
 - acts as the boundary between tiers
 - a tier must make a network call to interact with another tier
- Developing a multi-tier application - Application components:
 - Code that defines a message queue for communication between tiers
 - Code that defines an application programming interface (API) and a data model
 - Security-related code that ensures appropriate access to the application

Layers Vs Tiers

- Layer
 - Functional division of the software
- Tier
 - Functional division of the software that
 - runs on infrastructure separate from the other divisions
 - Presentation tier runs on one infra, but can have multiple layers
 - The Contacts app on your phone, for example, is a *three-layer* application, but a *single-tier* application, because all three layers run on your phone.

Process for developing n-tier Arch

- Problem: Automate the online selling of books to customers
- Step I: Define Main functions / purpose
- Step II: Define necessary business processes that require automation
- Step III: Translate business processes into requirements
- Step IV: Define Software arch
- Step V: Define number of server tiers

- [Google Cloud Arch](#)
- [VMWare cloud arch](#)